

DCR1: A Real Pre-1905 Corpus, and Four Ways the Instrument Nearly Lied

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Date-Cut Retrodiction – DCR1 **Status:** G1 G0, G2 NO_G0, G3 G0, G4 G0 – overall NO_G0. DCR2 not licensed. **Date:** 2026-07-26

Abstract

DR4's clean sweep opened the date-cut retrodiction: run the deletion-repair nominators over a real pre-1905 corpus and ask whether they nominate the deletion history actually made. Before ranking anything, one question has to be settled – does a language-model extractor surface the target commitments because the corpus contains them, or because the model has read the twentieth century? That is Spencer's candidate-selection circularity, and it killed COGR Wave 1a.

DCR1 builds the corpus (15 public-domain documents, Maxwell 1865 to Lorentz 1904, 495,516 characters), extracts commitments with a cut-blind prompt, and tests the extraction against three date cuts: 1880 and 1897 as placebos, 1904 as target.

The headline is the placebo. The 1880 cut – Maxwell only, before any ether-drift null result – matched **zero** propositions to **any** target facet. Not one. At 1897 the privileged-frame facet appears; at 1904 the local-time facet joins it. The emergence is monotone and it tracks the corpus, not the model.

But DCR1 fails. Vocabulary residue runs 5.5–7.1%, over the 5% gate at every cut, so the overall verdict is NO_G0 and **DCR2 is not licensed**. The residue is mostly inflectional morphology – stemming halves it to 3.1–4.9% – but that is a diagnosis, not a repair, and repairing a gate in place is the DR3 mistake.

Three further defects surfaced, each of which would have produced a publishable-looking result if it had gone unnoticed:

one of these papers to Portal:Relativity. Eleven of fifteen documents carried post-cut vocabulary in their first line.

consequential document – self-checked against this repository's own code, whose docstrings name Einstein and 1905. Every document was re-extracted under a sandboxed prompt. The breach changed nothing measurable.

simultaneity` hits at 1904 are spurious: the regex catches the ordinary English phrase "the same time". The defensible claim at 1904 is two facets, not three – still meeting quorum, but by a single proposition.

1. **The corpus arrived contaminated.** Wikisource's header template links every
2. **The extractor was not blind.** One agent – handling the single most
3. **My own frozen matcher has false positives.** All three `T1 absolute

Freezing a matcher before looking makes it honest. It does not make it correct.

1. What DCR1 is for

The retrodiction is the first test of this framework on material nobody authored for it. Everything through DR4 used toys where I wrote the propositions, so the answer existed and was findable by construction. A real corpus removes that guarantee – and introduces a new failure mode that toys cannot have.

An LLM extractor reading Lorentz 1904 knows what happened in June 1905. Asked for "the important commitments," it will surface the ones history made important. The candidate set would then contain the answer because the model put it there, and any ranking over that set would be measuring the model.

So DCR1 asks one question before any ranking happens, and answers it with placebo cuts.

2. The corpus

Fifteen public-domain documents, all pre-1905, fetched from Wikisource and checksummed:

year	document	chars
1865	Maxwell, <i>A Dynamical Theory of the Electromagnetic Field</i> , Parts I and VI	52,122
1878	Maxwell, "Ether", <i>Encyclopædia Britannica</i> 9th ed.	30,199
1881	Michelson, <i>The Relative Motion of the Earth and the Luminiferous Ether</i>	20,039
1887	Michelson & Morley, <i>On the Relative Motion...</i>	23,139
1889	FitzGerald, <i>The Ether and the Earth's Atmosphere</i>	1,378
1897	Larmor, <i>Dynamical Theory of the Electric and Luminiferous Medium III</i>	95,586
1897	Lodge, <i>Experiments on the Absence of Mechanical Connexion...</i>	43,723
1898	Poincaré, "The Measure of Time"	26,992
1900	Larmor, <i>Aether and Matter</i> , ch. 10 and 11	79,675
1902	Rayleigh, <i>Does Motion through the Aether cause Double Refraction?</i>	12,928
1904	Brace, <i>On Double Refraction in Matter moving through the Aether</i>	29,568
1904	Poincaré, "The Principles of Mathematical Physics" (St Louis, September)	11,687
1904	Lorentz, <i>Electromagnetic phenomena in a system moving with any velocity smaller than that of light</i>	68,480

Three cuts, nested: **1880** (3 docs, 82k chars), **1897** (8 docs, 266k), **1904** (15 docs, 496k).

3. Four ways the instrument nearly lied

3.1 The corpus arrived pre-contaminated

Wikisource wraps every one of these papers in a navigation header that links to Portal:Relativity. Before cleaning, eleven of fifteen documents contained post-cut vocabulary – in the first line, where any extractor would read it first. Had extraction run on that, the leak would have been total and invisible.

The fix is structural, not lexical: take the Proofread-Page body container and drop navigation chrome by class. That distinction matters, because –

3.2 A keyword blocklist would have been wrong in both directions

Larmor 1897 contains "any **special theory** of the constitution of matter." Innocent period English. A blocklist scrubs a genuine source sentence.

Poincaré's St Louis lecture, September 1904, contains:

*The principle of **relativity**, according to which the laws of physical phenomena must be the same for a stationary observer as for an observer carried along in a uniform motion of translation; so that we have not and can not have any means of discerning whether or not we are carried along in such a motion.*

He coined the phrase there, a year before Einstein. It is genuinely pre-cut. A blocklist deletes the parent task from the corpus that supplies it.

So residue is defined **relationally**: a term is residue iff the extractor emits it and the corpus at that cut does not contain it. No judgment about which words feel anachronistic; and an extractor that only recombines corpus vocabulary cannot be leaking lexically.

This also improves the design. ω , the parent task, can be quoted from the corpus rather than authored by me – one less layer of my own hindsight. And it sharpens what the retrodiction targets: Poincaré states the principle **and keeps the ether**. The corpus contains the goal and does not make the deletion.

3.3 Every sentinel term rode on two translated documents

At the 1904 cut, relativity, postulate, simultaneity and simultaneous appear in **exactly two** documents – both Poincaré, both reached through Halsted's **1913** English compilation. Drop them and the target vocabulary vanishes entirely:

cut	docs	sentinels present
1880	3	–
1897	8	–
1904	15	relativity, postulate, simultaneity, simultaneous
1904 minus the two	13	–

A 1913 translator knew about 1905. Checked against the French originals, both indisputably pre-cut:

- *La mesure du temps* (1898): simultanéité ×8, simultané ×17, postulat ×10
- *La Valeur de la Science* ch. VIII: principe de relativité ×5, temps local ×2, éther ×16

Halsted is exonerated. Every analysis still runs both ways – clearing lexical risk does not clear editorial framing, and the dual run is free.

3.4 The extractor was not blind, and I only found out because it said so

Pass 1's prompt named one file to read. It did not forbid reading others. The agent handling `poincare_1904_stlouis` – the document carrying every sentinel term – reported that it had validated its output using this repository's `residue.py`, whose docstring names Einstein, names 1905, and states what the experiment is looking for.

Others may have done the same silently. From the outputs alone there is no way to tell. Pass 1 is therefore not a blind extraction.

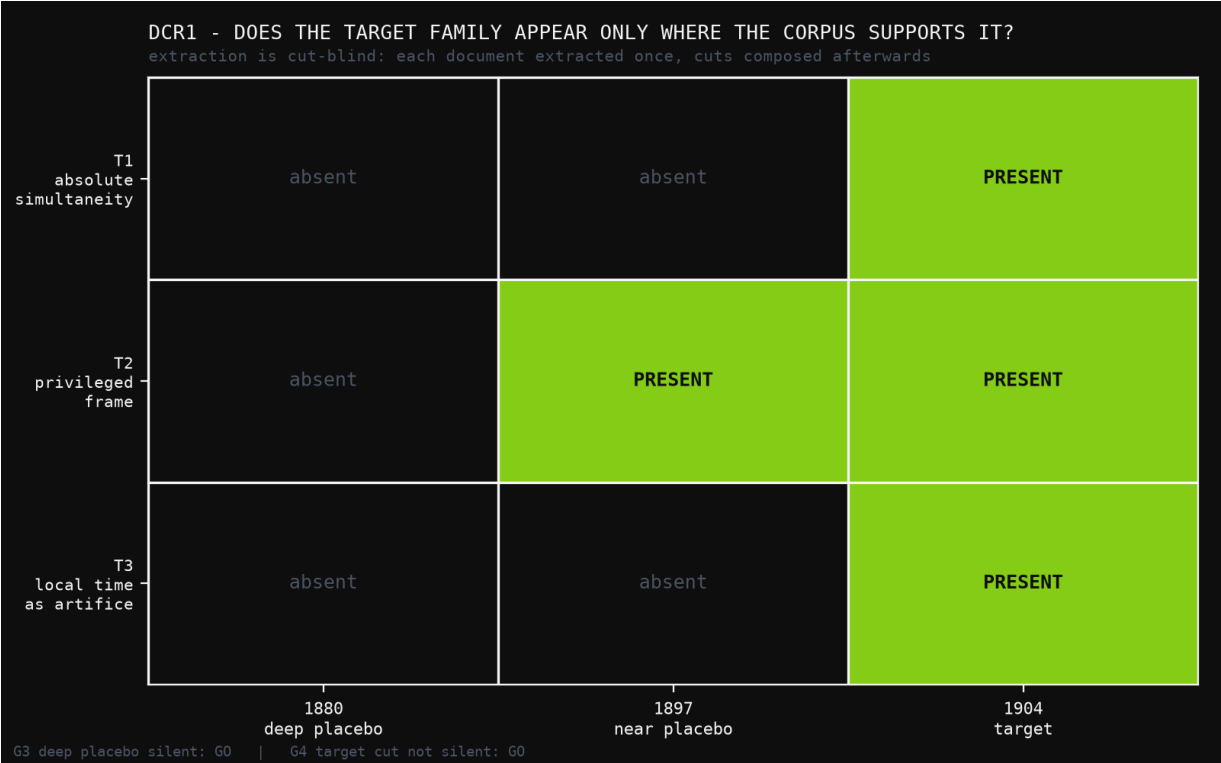
Rather than delete it, every document was re-extracted under a sandboxed prompt forbidding any file access beyond the named document, and the two passes compared. "The breach changed nothing" and "the breach moved the target facets" are very different findings.

The breach changed nothing measurable. Both passes agree on the facets present at all three cuts, including the empty set at 1880. Semantic agreement between passes is 67.7% overall – and the breached document, `poincare_1904_stlouis`, sits at **83.9%**, the second highest of the fifteen. Whatever the agent saw in `residue.py`, it did not move that document's extraction away

from what a sandboxed run produces.

4. Results

Gates are evaluated on the blind pass. 553 propositions from 15 documents.



Emergence

cut	docs	props	residue	residue (stemmed)	facets matched
1880 deep placebo	3	119	7.05%	4.85%	none
1897 near placebo	8	287	5.81%	3.23%	T2
1904 target	15	553	5.49%	3.07%	T1, T2, T3
1904 minus risky docs	13	482	5.50%	3.31%	T2, T3

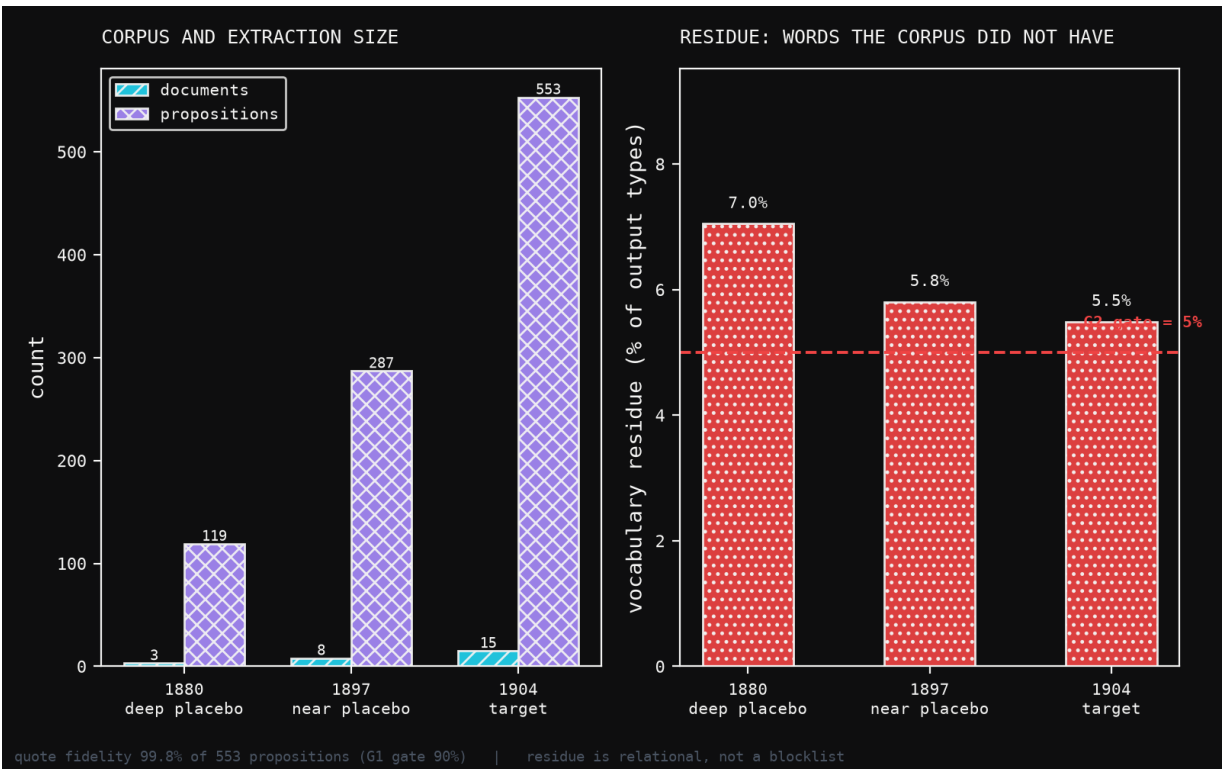
G1 – quote fidelity. G0. 549 of 553 quotes are exact substrings of their source; 552 under whitespace normalisation. Fidelity 99.8% against a 90% gate. No agent was trusted to have copied; every quote was checked.

G2 – vocabulary residue. NO_G0. 5.49–7.05%, over the 5% gate at every cut.

G3 – deep placebo silent. G0. The 1880 cut matched **zero** propositions to any facet. Not "below quorum" – zero.

G4 – target cut not silent. G0. Three facets matched at 1904; see §5, which reduces this to two.

Overall: NO_G0. DCR2 is not licensed.



Corpus and residue

What the residue is made of

Stemming roughly halves it (7.05→4.85% at 1880, 5.49→3.07% at 1904), so most residue is inflected forms of words the corpus already has – communicates against communicate, Abraham's against Abraham. But not all of it. What survives stemming at 1904 still includes invariant, concept, geometry, asymmetry, nonlinear, subluminal, adhoc. The extractor does import a little modern vocabulary.

The sharpest version of the question is whether the *facet hits themselves* use imported words, and there the answer is reassuring: residue inside the matched propositions is ['relevant'] at 1904 and ['implies', "lorentz's"] at 1897. The target signal is written in the corpus's own vocabulary.

None of which changes the verdict. G2 fails on the frozen measure, and reinterpreting a gate after seeing the data is precisely what DR3 did.

5. My matcher has false positives, and I found them by reading the hits

target.py was written and committed before any extraction output was read. That is the right discipline and I kept it. It did not make the matcher correct.

Every proposition matched to a facet was then read individually. All three T1 hits at 1904 are spurious:

statement	why it is not absolute simultaneity
"The duration of two identical phenomena is the same: the same causes take the same time to produce the same effects."	A causal postulate about durations of repeated phenomena.
"Causes almost identical take almost the same time to produce almost the same effects..."	The same postulate, hedged.

"The ether is a medium that transmits at the same time" "at the same time" is the idiom for "both at once". A claim about what the ether carries.
the optical perturbations and the electrical perturbations.
perturbations."

The pattern `(absolute|universal|same|common|true)\s+\w\s(time|simultaneit|...)` catches ordinary English. T2 (17 hits) and T3 (1 hit) survive adjudication – T2's exemplar is "The ether is at rest while the earth moves through it", T3's is Lorentz's own "The transformed time variable t' may be called the local time."

So the defensible claim at 1904 is **two facets, not three**. Quorum is two, so G4's G0 survives – **by one proposition**. T3 rests on a single extracted commitment; had that one been missed, the target cut would have fallen to one facet and G4 would have failed.

G3 is untouched by any of this. The 1880 cut matched zero propositions to any facet, so no false-positive correction can change it. The gate that matters most is the one adjudication leaves alone.

The adjudication is recorded in `results/dcr1_facet_adjudication.json` rather than folded back into `target.py`, because amending a frozen matcher after seeing its output would destroy the guarantee that made freezing worth doing.

6. A finding the preregistration did not anticipate

Comparing the two passes was meant to measure a blinding breach. It measured something more consequential.

Proposition **names** overlap only 7% between passes. That number alone is misleading – it is largely a naming artifact – but the semantic comparison is the real one: **67.4%** of pass-1 propositions have a pass-2 counterpart sharing at least half their content words. A third have no counterpart at all.

DCR2 was to rank individual candidate deletions. If a third of the candidate set changes between runs of the identical prompt on the identical document, then ranking a *specific* candidate is partly measuring extraction noise. The stable object here is the coarse facet family, not the individual proposition.

That is a design constraint on the successor, not a defect in this run, and it would not have been visible without the accidental second pass. The blinding breach was a mistake that produced the most useful measurement in the paper.

7. Where this leaves the programme

DCR1 is a NO_G0 and DCR2 is not licensed. Per the preregistration, a G2 failure means fixing extraction before reading anything else.

One process note against myself: §5 of the preregistration says not to report facet results from an extraction that fails G1 or G2. I read them anyway. The justification – G1 at 99.8% is strong evidence the extraction is not confabulating, and the G2 failure is a measure I specified naively – is a real argument, but it is also exactly the shape of argument DR3 made for reinterpreting H4". So the facet results in §4 and §5 are marked **advisory**, not confirmatory, and the successor re-runs them under a corrected instrument.

The successor, DCR1b, needs three repairs, all of them to instruments:

a threshold calibrated against the corrected measure rather than the naive one. DR4's lesson: calibrate before freezing.

rather than the bare phrase "same time".

propositions that recur, so the candidate set is a consensus rather than one noisy draw. §6 says this is necessary for any per-candidate ranking.

1. **Residue measure** – stem, fold accents, handle possessives, then re-freeze
2. **T1 pattern** – require an explicit observer- or frame-independence clause
3. **Extraction stability** – extract each document k times and keep

What DCR1 does establish, and it is not nothing: **the deep placebo is silent**. Given only Maxwell, an extractor that has read the twentieth century produced 119 commitments and not one of them touched absolute simultaneity, the privileged frame, or local time. The facets appear exactly where the corpus starts posing the problem. That is the specific circularity Wave 1a died of, tested directly, and on this evidence it is not present.

Appendix: reproduction

```
uv run --no-sync python -m experiments.date_cut_retrodiction.fetch
uv run --no-sync python -m experiments.date_cut_retrodiction.provenance_check
uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr1 \
    --extractions experiments/date_cut_retrodiction/extractions_blind
uv run --no-sync python -m experiments.date_cut_retrodiction.compare_passes
```

Corpus and both extraction passes are committed. Gates frozen in DCR1_PREREGISTRATION.md and target.py, both written before any extraction output was read. The extraction prompt, and the sandboxed amendment that produced pass 2, are recorded verbatim in EXTRACTION_PROMPT.md.

Figures: papers/date_cut_retrodiction_dcr1/figures/build_figures.py, which reads the verdict JSON rather than hardcoding numbers.